

Optimized joint LiFi coordinated multipoint joint transmission clustering and load balancing for hybrid LiFi and WiFi networks

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Hybrid light fidelity (LiFi) and wireless fidelity (WiFi) networks provide a promising solution for future indoor wireless communication. Hybrid networks face the challenge of imbalanced cell load due to the uneven distribution of user equipment (UE), and they need to suppress inter-cell interference induced by the dense deployment of LiFi access points (APs). In this paper, with the aim to efficiently establish UE-AP associations and allocate resources, we propose a potential game (PG) aided design of joint user-centric LiFi coordinated multipoint (CoMP) joint transmission (JT) clustering and load balancing (LB). In our design, we formulate the joint design problem as a mixed-integer nonlinear programming (MINLP) problem, which integrates the requirement for the minimum data rate of each UE. With the aim to maximize the system throughput under proportional fairness constraints, we propose to build a PG model to solve the MINLP problem. Then we derive the Nash equilibrium based on the best-response dynamics, and thus a suboptimal solution is attained. Thanks to the space diversity gain brought by our joint LiFi CoMP-JT clustering and LB design, the signal-to-interference-plus-noise ratio can be improved. Thus, when the physical layer configurations remain the same, hybrid networks can achieve larger capacity and better reliability performances. Simulation results validate that hybrid networks using our presented design would achieve better throughput and satisfaction performances than benchmarks. © 2020 Optical Society of America

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1. INTRODUCTION

Light fidelity (LiFi) has attracted considerable attention recently, as the vast unregulated spectrum of visible light (VL) enables a great potential to provide high communication capacity [1]. (This paper is presented in part in IEEE VTC-Fall 2019 [2]). However, stand-alone LiFi networks might exhibit some drawbacks such as poor non-line-of-sight (NLoS) performances due to the line-of-sight (LoS) transmission nature of VL. Therefore, heterogeneous networks are developed that integrate traditional radio frequency (RF) technologies such as wireless fidelity (WiFi) with LiFi networks to provide complementary services by exploiting the diffraction capabilities of RF propagation. In this context, hybrid LiFi and WiFi networks are regarded as a promising solution to meet the increasing wireless communications demands, especially in indoor scenarios [3–9].

The heterogeneous deployment brings the issue of imbalanced traffic load. The user equipment (UE) is usually unevenly distributed in hybrid networks. When the received-signal-strength (RSS)-based access point (AP) selection strategy

is used, most UEs would be attracted to the AP that provides the most powerful link, and thus traffic loads across the APs are greatly different, thereby bringing inefficient resource utilizations [3,4,9]. As a result, some APs are overloaded while the others are lightly loaded.

In addition, due to the LoS transmission over the VL band, each AP only covers a small area, referred to as the attocell, due to the large propagation attenuations over the VL band [1]. Thus, APs have to be densely deployed to provide services, inducing the inter-cell interference (ICI) issue [10,11], which leads to performance degradations of a smaller signal-to-interference-plus-noise ratio (SINR), lower data rate, etc. [12], especially for cell-edge users. Although [13,14] have optimized the deployment of positions of light-emitting diodes (LEDs) to improve the system performances, such as the achievable rate and the imbalanced traffic load, ICI might still degrade the performances.

Naturally, in order to improve the system performance and ensure user fairness, both *imbalanced traffic load* and *ICI* issues should be addressed for hybrid LiFi and WiFi networks.

A. Related Works

On the one hand, coordinated multipoint (CoMP) joint transmission (JT) schemes are proposed to combat the LiFi ICI by making use of shared information among cooperative APs. CoMP-JT utilizes multiple APs to cooperatively transmit the UE's data at the same time-frequency resource and thereby converts interfering signals into useful signals. CoMP-JT schemes can achieve considerable gains in system throughput [15], robustness to blockage [11], and signal coverage [16]. Compared with RF networks, it is easier to implement CoMP-JT in LiFi systems, since the use of intensity modulation/direct detection (IM/DD) makes it easier to synchronize the signals and combat the multipath fading effect. Additionally, the indoor deployment of LiFi systems enables an ideal backhaul solution of power-line communication (PLC) to support coordination and data exchange among cooperating APs in CoMP-JT operations [16].

Besides, compared with other ICI mitigation schemes, such as frequency reuse, by referring to [12] we can see that although the frequency reuse scheme is simpler to implement, the bandwidth efficiency is lower, and the user demands are not considered. By contrast, the CoMP method is adjustable to any AP arrangement with the aid of the central control unit (CCU) while taking the user information into account. Hence, in this design and in our previous work [2], we propose to apply the clustering scheme to suppress the ICI.

On the other hand, load balancing (LB) schemes are developed to address the imbalanced cell load issue by transferring some traffic loads from busy APs to idle APs. Due to different propagation characteristics of LiFi and WiFi links, LB schemes developed for conventional heterogeneous RF networks cannot be directly applied. Thus, LB schemes are proposed for hybrid LiFi and WiFi networks to combat the shadowing effect [3], to accommodate the UEs' quality of experience (QoE) demands for the data rate [4], to establish cooperative formations of LiFi cells [5], or to reduce the handover overhead [6,7]. With the efficient UE-AP association and optimized resource allocations, these presented schemes achieve satisfactory performance gains.

However, few research works have been done on the traffic load allocation issue for hybrid CoMP-JT LiFi and WiFi networks. Different from conventional hybrid LiFi and WiFi networks, whose network topology is static and remains invariant, in the considered hybrid CoMP-JT and WiFi networks, the network topology will be dynamically changing due to the variation of clustering strategies, which brings a challenge for the LB design. Since the LB decisions are dependent on the topology [17], in the considered hybrid CoMP-JT and WiFi networks, in order to address the issues of both the *imbalanced traffic load* and *LiFi ICI* simultaneously, we propose to jointly design the LB and LiFi CoMP-JT clustering strategies.

B. Our Design

Different from our previous work presented in [2], in this paper, we address the *imbalanced cell load* and *LiFi ICI* issues, which are correlated with each other. Different from the above-mentioned schemes, a hybrid CoMP-JT LiFi and WiFi

network is considered, and we propose a joint design of user-centric LiFi CoMP-JT clustering and LB in order to determine the efficient UE-AP association and time resources allocation with an objective of maximizing the system throughput under proportional fairness constraints.

More explicitly, taking different UE's QoE demands for the data rate into account in our design, we exploit the complementary advantages of LiFi and WiFi networks to serve the UEs. Namely, in our design, the LiFi network would make the best efforts to provide higher data rate transmissions by utilizing the broad VL band, while the WiFi AP would provide access for users to meet their minimum data rate requirements by using the RF diffraction capability. In this context, we formulate the joint design of user-centric LiFi CoMP-JT clustering and LB as a mixed-integer nonlinear programming (MINLP) problem with considerations of each UE's requirement for the minimum data rate, with the aim to find a balanced UE-AP association. Since the couplings of decision variables brings high computation complexity for problem solving, we propose to apply game theory to find a stable suboptimal solution.

Different from the existing modeling methods, such as the evolutionary game in [3], we build up a generalized ordinal potential game (GOPG) model, wherein the Nash equilibrium (NE) is found based on best-response (BR) dynamics based on the defined strategy set. Then we analyze the performances of the proposed scheme via simulations in terms of the system throughput, the user satisfaction, the UE QoE index distribution, the SINR, and the outage rate, which outperform the single-point transmission (SPT) schemes and the combined transmission (CT) scheme proposed in [5].

Notably, the main contributions of this paper include

1. Considering the dynamically changing network topology of indoor hybrid CoMP-JT LiFi and WiFi networks, which is distinct from the static network topology of traditional hybrid LiFi and WiFi networks, we propose to jointly design the LB and the user-centric CoMP-JT clustering strategy. Thus, the resources can be allocated efficiently while suppressing the ICI.
2. We exploit different propagation characteristics of LiFi and WiFi links to serve the UEs, i.e., we utilize LiFi APs to provide high-speed links while using the WiFi AP to meet the UEs' QoE requirements on the minimum data rate. Then the joint resource optimization problem is formulated as a MINLP problem with considerations of each UE's requirements for the minimum data rate. Moreover, our objective is to find a balanced UE-AP association and time slot resource allocations.
3. We construct the strategy set and build a game model, which is proved to be a GOPG. Then we derive the NE based on BR dynamics as a suboptimal solution to the formulated MINLP problem. By exploiting the space diversity gain brought by the CoMP-JT, the SINR can be improved, thereby leading to higher data rate and improved reliability performances when the physical layer configurations remain the same. Simulation results are provided to validate our design in the deployment of hybrid CoMP-JT LiFi and WiFi networks.

The rest of the paper is organized as follows. Section 2 introduces the general system model of hybrid networks. Section 3 presents the problem of joint LiFi CoMP-JT clustering and LB; then, based on game theory, we describe the game model and provide solutions for the joint design. Section 4 provides simulations to verify the effectiveness of the proposed scheme. Finally, Section 5 concludes the paper.

2. SYSTEM MODEL

A. System Setup

Figure 1 illustrates a generalized indoor hybrid LiFi and WiFi system model. With considerations of the implementation complexity and referring to [3,5,6], on the ceiling, we uniformly place multiple LiFi APs with unity frequency reuse (UFR). Then, in order to utilize the wide coverage of the WiFi AP equally in each corner of the room, we place the WiFi AP at the center of the ceiling [3,8,9]. It is worth mentioning that with the usage of the position planning proposed by [13,14], the system performance might be further enhanced.

Notably, as pointed out by [18], in order to keep the CoMP-JT implementable, it is required that the signals from multiple APs arrive at the receiver with latency smaller than a symbol time. In our considered indoor scenarios, we assume that all APs are wired to the CCU via optical fibers or PLC [3,6,11] to shorten the time delay induced by the transmissions. Namely, without the loss of generality, we assume that the channel-state information (CSI) estimation time and the transmission time delay are small and negligible to support the implementation of CoMP-JT. Besides, a set of UEs is randomly distributed in the system. Each one is equipped with a LiFi receiver and a WiFi receiver, and each can access one kind of network (LiFi or WiFi) at a time with considerations of the hardware implementation complexity [3,5–8]. The sets of LiFi APs, WiFi APs, and UEs are respectively denoted as $\mathbf{L}$, $\mathbf{W}$, and $\mathbf{U}$.

Considering that the LiFi subsystem relies on the infrared-based technique for uplink, while the WiFi subsystem operates in the time-division duplex (TDD) mode, the interference of uplink transmissions exerted on the downlinks is very limited. Therefore, without the loss generality, and with reference to [3–8,11,15], we neglect the impacts of the uplinks on RF or VL channels in the following discussions. In addition, we consider a full-buffer traffic model, and the time-division multiple access (TDMA) scheme is used by both the LiFi and WiFi APs. To be more explicit, a time interval is divided into multiple slots, and the UEs share the same frequency channel but consume the time slot separately [19]. Thus, we use η_u^{LiFi}

and η_u^{WiFi} to denote the portion of the time resource allocated to UE $u \in \mathbf{U}$ in the LiFi and WiFi subsystems, respectively, which have a value between 0 and 1 according to [3,6,7].

In the considered indoor scenario, with loss of generality, we assume the UE is static or moved by walking. Thus, the channel condition will change slowly, which means that the CSI collected by the CCU would remain invariant in the time durations of carrying out the clustering and LB balancing strategy [3,5]. The CCU will allocate the resources periodically. Thus, in the short duration of one period, the mobility of the UE has limited impact on the performances and hence is negligible in our following discussions.

In the LiFi subsystem, due to the LoS propagation characteristic of VL channels, unlike the traditional cellular system, the LiFi APs are not clustered by sectorizing cells and reforming them. Instead, in our design, LiFi APs are grouped into clusters according to the user demands, and the clustering is dynamically changing, i.e., we adopt the user-centric clustering strategy. With reference to [17,20] and considering the trade-off between performance gain and the cost including the signaling overhead and the scheduling complexity, we propose to use $l_{\max}$ LiFi APs with the strongest RSS as candidate APs for clustering, where $l_{\max}$ is the allowed the maximum cluster size and the set of candidate APs for UE u is denoted as $\mathbf{L}_u$. Notably, the cluster size can be 1, 2, ..., $l_{\max}$, and when $l_{\max} = 1$, SPT will be performed. Then, for the UE u in a specific cluster, it will be served by multiple LiFi APs, which transmit the same signal with the same time-frequency resource. In this way, the received power increases via the larger space diversity gain, and the ICI can be suppressed. In addition, it is worth mentioning that in the cluster, thanks to the independent distribution of APs connected via optical fibers or PLC, and the constrained coverage due to LoS propagation characteristics, the co-channel interference inside a sector for the nonnegative optical signals could be neglected, and the precoding would not be required to be applied in our system model.

In the considered system, at the beginning of each time interval, the UEs estimate the CSI for both LiFi and WiFi links and report the CSI via uplinks to the CCU [6]. Without loss of generality, we assume that perfect CSI could be learned. Then, based on the collected CSI, the CCU can execute the joint LB and CoMP-JT clustering algorithm to provide optimized UE-AP association decisions and time resource allocation [3,6,11]. Namely, in the considered system, the CCU will determine how to jointly formulate the cluster and serve the UE with multiple APs. Each LiFi AP would receive the instruction from the CCU and learn that it should serve a specific user.

After the UE-AP association is determined, if more LiFi APs serve the UE via CoMP-JT, the SINR will become larger accordingly. Notably, no localization subsystem is required. Next, we will derive the achievable data rate as follows.

B. Achievable Data Rate

1. LiFi Channel

For LiFi channels [5,9], we neglect NLoS components and only consider LoS components. For UE $u \in \mathbf{U}$ and LiFi AP $l \in \mathbf{L}$, let $d_{u,l}$, $\phi_{u,l}$, and $\psi_{u,l}$, respectively, denote the Euclidean

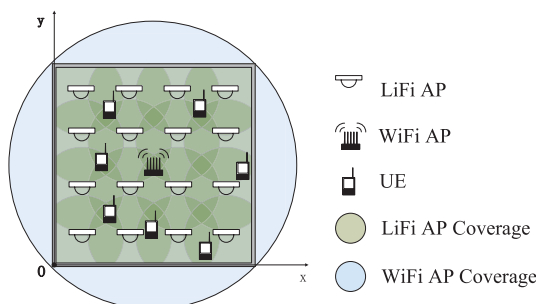


Fig. 1. Top view of an indoor hybrid LiFi and WiFi system.

distance, the angle of irradiance, and the angle of incidence of the LoS path. The LiFi channel gain is denoted by [21]

$$H_{u,l}^{\text{LiFi}} = \frac{(m+1)A_{\text{pd}}}{2\pi d_{u,l}^2} \cos^m(\phi_{u,l}) g_f g_c(\psi_{u,l}) \cos(\psi_{u,l}), \quad (1)$$

where $m = -\ln 2 / \ln(\cos(\Phi_{1/2}))$ is the Lambertian emission order and $\Phi_{1/2}$ is the radiation angle at half of the illuminance of the source. A_{pd} is the physical area of the photodetector, g_f is the gain of the optical filter, and $g_c(\psi_{u,l})$ is the optical concentrator gain, whose value is given by

$$g_c(\psi_{u,l}) = \begin{cases} \frac{n^2}{\sin^2(\Psi_{\max})}, & 0 \leq \psi_{u,l} \leq \Psi_{\max} \\ 0, & \psi_{u,l} > \Psi_{\max} \end{cases}, \quad (2)$$

where n is the refractive index and $\Psi_{\max}$ is the semi-angle of the field of the view (FoV) of the photodetector.

Let binary variable $\alpha_{u,l}$ denote the association state between UE u and LiFi AP l , which has a value of 1 for the association state and 0 otherwise. Then $\alpha_u = \{\alpha_{u,l} | l \in \mathcal{L}\}$ would denote the cluster decision for UE u . It is worth mentioning that the two kinds of JT schemes, i.e., coherent (C-) and non-coherent (NC-) JTs [16,20], are compatible with the proposed joint design. Then, for the user $u \in \mathcal{U}$ and the clustering decision α_u , the SINR expressions when applying C-JT or NC-JT can be obtained as [16]

$$\gamma_u^{\text{LiFi,C-JT}}(\alpha_u) = \frac{(\sum_{l \in \mathcal{L}} \alpha_{u,l} r_{\text{pd}} H_{u,l}^{\text{LiFi}} P_t^{\text{LiFi}} / \kappa)^2}{N^{\text{LiFi}} B^{\text{LiFi}} + \sum_{l \in \mathcal{L}} (1 - \alpha_{u,l}) (r_{\text{pd}} H_{u,l}^{\text{LiFi}} P_t^{\text{LiFi}} / \kappa)^2}, \quad (3)$$

$$\gamma_u^{\text{LiFi,NC-JT}}(\alpha_u) = \frac{\sum_{l \in \mathcal{L}} \alpha_{u,l} (r_{\text{pd}} H_{u,l}^{\text{LiFi}} P_t^{\text{LiFi}} / \kappa)^2}{N^{\text{LiFi}} B^{\text{LiFi}} + \sum_{l \in \mathcal{L}} (1 - \alpha_{u,l}) (r_{\text{pd}} H_{u,l}^{\text{LiFi}} P_t^{\text{LiFi}} / \kappa)^2}, \quad (4)$$

where r_{pd} denotes the detector responsivity, P_t^{LiFi} denotes the transmitted power, κ denotes the conversion coefficient such that $P_t^{\text{LiFi}} / \kappa$ is equivalent to the optical signal power [8], N^{LiFi} denotes the power spectral density (PSD) of noise at the detector, and B^{LiFi} denotes the bandwidth of the LiFi AP.

Notably, if more LiFi APs are involved in the CoMP-JT clustering decision for UE u , $\alpha_{u,l}$ will have a larger value. Then, from the SINR expression presented by Eqs. (3) and (4), we could observe that the numerators will become greater, while the denominators will become smaller. As a result, the SINR will increase and have a larger value.

In the LiFi subsystem, we assume that the M -ary pulse amplitude modulation (M-PAM) scheme is applied, while the modulation order remains invariant during the AP selection procedure and the information transmissions. Then the achievable rate of UE u with the clustering decision α_u can be accordingly expressed as [22]

$$C_u^{\text{LiFi,C-JT}}(\alpha_u) = \frac{B^{\text{LiFi}}}{2} \log_2 \left[1 + \frac{\gamma_u^{\text{LiFi,C-JT}}(\alpha_u)}{\Gamma(M, P_e)} \right], \quad (5)$$

$$C_u^{\text{LiFi,NC-JT}}(\alpha_u) = \frac{B^{\text{LiFi}}}{2} \log_2 \left[1 + \frac{\gamma_u^{\text{LiFi,NC-JT}}(\alpha_u)}{\Gamma(M, P_e)} \right], \quad (6)$$

where P_e is the preset symbol error probability and Γ is the SINR gap given by [22]

$$\Gamma(M, P_e) = \frac{1}{3} \left[Q^{-1} \left(\frac{MP_e}{2(M-1)} \right) \right]^2, \quad (7)$$

where $Q(x)$ is the Q function.

For the sake of brevity, we neglect C-JT and NC-JT in notations, since the following derivations are suitable to be used for both kinds of JTs.

2. WiFi Channel

For WiFi channels, the TGn channel model [23,24] is considered. Denote the distance between UE u and the WiFi AP as d_u , and the path loss of the WiFi channel is written as

$$L(d_u) = \begin{cases} L_{\text{FS}}(d_u) + X_\sigma, & d_u \leq d_{\text{BP}} \\ L_{\text{FS}}(d_u) + 35 \log_{10}(\frac{d_u}{d_{\text{BP}}}) + X_\sigma, & d_u > d_{\text{BP}} \end{cases}, \quad (8)$$

where $L_{\text{FS}}(d_u) = 20 \log_{10}(d_u) + 20 \log_{10}(f_c) - 147.5$ is the free-space loss, f_c is the central carrier frequency, d_{BP} is the breakpoint distance, and X_σ is a Gaussian random variable with zero mean and standard deviation σ . The multipath propagation of the WiFi channel is formulated as

$$\rho_u^{\text{WiFi}} = \sqrt{\frac{K}{K+1}} e^{j\phi_u} + \sqrt{\frac{1}{K+1}} X_1, \quad (9)$$

where K is the Ricean K-factor, ϕ_u is the arrival angle of the LoS path, and X_1 is a Gaussian random variable with zero mean and unit variance.

Considering that only one WiFi AP exists, we neglect the RF interference. The SINR of UE u is expressed as

$$\gamma_u^{\text{WiFi}} = \frac{|\rho_u^{\text{WiFi}}|^2 10^{-L(d_u)/10} P_t^{\text{WiFi}}}{N^{\text{WiFi}} B^{\text{WiFi}}}, \quad (10)$$

where P_t^{WiFi} is the transmitted power, N^{WiFi} is the noise PSD at the receiver, and B^{WiFi} is the bandwidth consumed by the WiFi AP.

In the WiFi subsystem, we assume that the M -ary quadrature amplitude modulation (M-QAM) scheme is adopted, while the modulation order also remains invariant during the AP selection and the information transmissions. The achievable WiFi rate is thereby expressed as [25]

$$C_u^{\text{WiFi}} = B^{\text{WiFi}} \log_2 \left[1 + \frac{\gamma_u^{\text{WiFi}}}{\Gamma'(P_e)} \right], \quad (11)$$

where Γ' is the SINR gap given by [25]

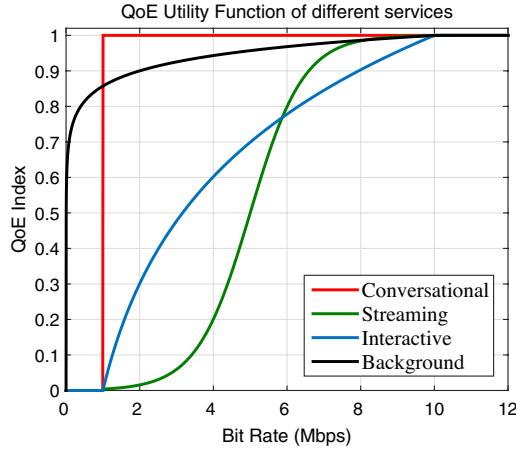
$$\Gamma'(P_e) = \frac{1}{3} \left[Q^{-1} \left(\frac{P_e}{4} \right) \right]^2. \quad (12)$$

C. Service Types and QoE

QoE is a concept reflecting the users' experience of the system performances, including the data rate, delay, jitter, or their combinations, which is related to but different from the quality of service (QoS) [26]. More explicitly, as pointed out in [27],

Table 1. Utility Functions for Services of Different Types [32]

Service	Utility Function	Parameter
Conversational	$\mathcal{U}_C(r) = \frac{\text{sgn}(r - R_{C,\min}) + 1}{2}$	$R_{C,\min} = 1 \text{ Mbps}$
Streaming	$\mathcal{U}_S(r) = \frac{r^2}{1 + (\frac{1}{\epsilon} - 1)e^{-rR_S}}$	$R_{S,\max} = 10 \text{ Mbps}$, $\epsilon = 0.001$
Interactive	$\mathcal{U}_I(r) = \frac{2 \ln(\frac{1}{\epsilon} - 1)}{R_{S,\max}}$	$R_{I,\min} = 1 \text{ Mbps}$, $R_{I,\max} = 10 \text{ Mbps}$
Background	$\mathcal{U}_B(r) = \frac{\log(r+1)}{\log(R_{B,\max}+1)}$	$R_{B,\max} = 10 \text{ Mbps}$

**Fig. 2.** QoE index of different services versus the data rate.

QoS is defined from a system's perspective, whereas QoE is defined entirely from the user's perspective. Besides, QoS does not take into account the user perceived quality and the degree of satisfaction, whereas QoE does. In the literature, [28–31] have proposed different utility functions to quantify the relationship between user satisfaction degree and the data rate, based on which QoE-aware designs are developed.

In this paper, we adopt the utility functions proposed in [32] to evaluate the degree of user satisfaction from the user's perspective. The utility functions quantify the degree of user satisfaction under different data rates. The quantified satisfaction degree is referred to as the QoE index, which has a value between 0 and 1, and a larger value represents a higher satisfaction degree. Here we consider four types of services, which are conversational, streaming, interactive, and background services. Table 1 summarizes the utility functions of different service types, where $\text{sgn}(\cdot)$ is the sign function, $R_{i,\min}$ is the data rate of service i below which the QoE index is 0, $R_{i,\max}$ is the data rate of service i at which the QoE index reaches 1, and ϵ is a constant. Figure 2 plots the utility functions, from which we can see that QoE indexes increase nonlinearly with the data rate and then become saturated when the rate increases to a threshold value.

Denote the QoE requirement of UE u for the i th kind of service as $Q_{u,i}$, whose value is between 0 and 1. Then the minimum data rate to satisfy this UE is $\mathcal{U}_i^{-1}(Q_{u,i})$, where $\mathcal{U}_i(\cdot)$ is the utility function of the i th type of service and $\mathcal{U}^{-1}(\cdot)$ denotes the inverse function of $\mathcal{U}(\cdot)$. Notably, we only discuss

the inverse function of the utility functions in their strongly increasing regions. For the conversational service, we define that $\mathcal{U}_C^{-1}(Q_{u,i}) = R_{C,\min}$.

3. JOINT DESIGN OF USER-CENTRIC LiFi COMP-JT CLUSTERING AND LB

In this section, we first formulate the problem of joint user-centric LiFi CoMP-JT clustering and LB. Subsequently, we build a game model to derive the solution for the joint design and to achieve NE based on the BR dynamics.

A. Problem Formulation

1. Time Resource Allocation and UE Throughput

In hybrid networks, the LiFi and WiFi APs have different link propagation characteristics and available bandwidth, which is taken into account in our LB design. More explicitly, the LiFi AP may consume much more bandwidth than the WiFi AP, e.g., the LiFi APs in our simulations have a system bandwidth of 40 MHz, while the WiFi AP has a bandwidth of 20 MHz, which indicates that the LiFi APs are better at providing high-data-rate communication links. However, due to the LoS propagations over VL bands, LiFi signals might be blocked by obstacles. By contrast, RF signals are capable of NLoS transmission thanks to their diffraction nature, and thus the WiFi AP can provide wider coverage and meet the minimum data rate requirements for the UEs distributed in the whole indoor scenario. Therefore, in our design, we propose to utilize the different advantages brought by the LiFi AP and WiFi AP, thereby achieving a higher user satisfaction degree for hybrid indoor systems. Thus, given the WiFi channel capacity C_u^{WiFi} and QoE requirement $Q_{u,i}$ of UE u , the ratio of allocated time slot resources to satisfy this UE is determined by

$$\eta_u^{\text{WiFi}}(\beta_u, Q_{u,i}) = \beta_u \mathcal{U}_i^{-1}(Q_{u,i}) / C_u^{\text{WiFi}}, \quad (13)$$

where β_u is a binary variable representing the association state between UE u and the WiFi AP, whose value is 1 for association and 0 otherwise.

For LiFi APs, CoMP-JT requires multiple APs in one cluster to allocate the same time-frequency resources to a specific UE. Recall that in SPT [33], given the UE-to-LiFi-AP association $\alpha = \{\alpha_u | u \in \mathcal{U}\}$, the unique optimal resource allocation that maximizes the system throughput under proportional fairness constraints would be $\alpha_{u,l} / N_l$, $\forall u \in \mathcal{U}$, $\forall l \in \mathcal{L}$, where $N_l = \sum_{u \in \mathcal{U}} \alpha_{u,l}$ denotes the number of UEs associated with LiFi AP l . That is to say, the UEs served by the same AP share the same resources.

Since we consider user-centric CoMP-JT clustering, multiple LiFi APs in the same cluster may serve a different number of UEs. Thus, we apply the following “load-aware” principle [17] for resource allocation:

$$\eta_u^{\text{LiFi}}(\alpha) = 1 / N_u^{\text{LiFi}}, N_u^{\text{LiFi}} = \max_{\alpha_{u,l}=1, l \in \mathcal{L}} N_l, \quad (14)$$

where N_u^{LiFi} is the maximum of N_l and l refers to the LiFi APs in UE u 's cluster. Here is an example to explain Eq. (14): Consider that a clustering decision for UE n is $\alpha_n = \{\alpha_{n,1}, \alpha_{n,2}, \dots, \alpha_{n,|L|}\} = \{1, 1, 1, 0, \dots, 0\}$ when LiFi

APs 1, 2, and 3 respectively have $N_1 = 8$, $N_2 = 4$, and $N_3 = 5$ UEs to serve. Then, according to Eq. (14), the fraction of resources for UE n would be $\eta_n^{\text{LiFi}} = 1/8$, since there is $N_n^{\text{LiFi}} = \max\{N_1, N_2, N_3\} = 8$.

Then we can evaluate the throughput of UE u as

$$R_u(\alpha, \beta_u, Q_{u,i}) = \eta_u^{\text{LiFi}}(\alpha) C_u^{\text{LiFi}}(\alpha) + \eta_u^{\text{WiFi}}(\beta_u, Q_{u,i}) C_u^{\text{WiFi}}. \quad (15)$$

2. Joint Design Problem

With the objective of maximizing the system throughput under proportional fairness constraints, the logarithmic function in the form of $\log(R_u)$ [6] is adopted in the objective function. Then we formulate the joint resource allocation problem as

$$\begin{aligned} \max_{\alpha, \beta} \quad & \sum_{u \in U} \log(R_u(\alpha, \beta_u, Q_{u,i})) \\ \text{s.t.} \quad & C1: \sum_{l \in L} \alpha_{u,l} \leq l_{\max} \quad \forall u \in U; \\ & C2: \sum_{u \in U} \beta_u \eta_u^{\text{WiFi}}(\beta_u, Q_{u,i}) \leq \eta_{\max}^{\text{WiFi}}; \\ & C3: \mathbb{I}(\sum_{l \in L} \alpha_{u,l} > 0) + \beta_u = 1 \quad \forall u \in U; \\ & C4: \eta_u^{\text{LiFi}}(\alpha), \eta_u^{\text{WiFi}}(\beta_u, Q_{u,i}) \in [0, 1] \quad \forall u \in U; \\ & C5: \alpha_{u,l}, \beta_u \in \{0, 1\} \quad \forall u \in U, \quad \forall l \in L; \end{aligned} \quad (16)$$

where the decision variables α and β are association indicators. In addition, we have $\beta = \{\beta_u | u \in U\}$. For the sake of simplicity, we neglect the random variables in the expressions, e.g., $\eta_u^{\text{LiFi}}(\alpha)$ is simplified as η_u^{LiFi} .

Notably, in Eq. (16), constraint C1 means a LiFi CoMP cluster has a limited size, which should be smaller than $l_{\max}$; constraint C2 ensures that the occupied WiFi resources' fraction is less than $\eta_{\max}^{\text{WiFi}}$, where $\eta_{\max}^{\text{WiFi}}$ is the downlink resource quota allocated by the WiFi AP, while the remaining time fraction of $1 - \eta_{\max}^{\text{WiFi}}$ might be used for uplinks [5]; constraint C3 means that each UE is connected to either the LiFi network or the WiFi network, where $\mathbb{I}(A)$ is the indicator function that maps to 1 when A is true and 0 otherwise; constraint C4 sets the range of the time resource fractions allocated to each UE; and constraint C5 means that the decision variables are all binary.

Naturally, Eq. (16) is a MINLP problem. The complexity of brute-force search is $O((2^{l_{\max}} C(l_{\max}, N_L))^{N_U})$, where N_L and N_U are the numbers of LiFi APs and UEs, respectively, and $C(a, b)$ is the combination operation. The complexity is very high, even for modest-sized hybrid networks. Besides, based on Eq. (3) and Eq. (14), we can see that decision variables $\alpha_{u,l}$ couple with each other in the objective function, making it infeasible to implement the Lagrange dual decomposition method [5]. With the aim to solve the MINLP problem with low complexity, we propose to use game theory to derive the optimal resource allocation strategy, since UEs in hybrid networks are autonomous players competing for limited resources and their interests conflict.

B. Game Model Buildup

The game $\mathbb{G}$ is formulated as follows:

- 1) Player Set U : The players of the game are the UEs in an indoor scenario.

- 2) Strategy Set $\{\mathcal{S}_u\}_{u \in U}$: For the considered hybrid CoMP-JT LiFi and WiFi networks, as mentioned above, in our design, not only UE connections to different access networks but also the user-centric CoMP-JT clustering choices are jointly taken into account in the resource allocation. Naturally, the strategy set is different from other game-theory-based solutions presented for specific systems. Namely, the strategy set in the game formulation should be constructed according to the specific network operation conditions. We here propose to construct the strategy set for UE $u \in U$ as $\mathcal{S}_u = \{S_0, S_1, S_2, \dots, S_{2^{l_{\max}}-1}\}$, where S_0 is the strategy of connecting to the WiFi AP, and the other $2^{l_{\max}} - 1$ strategies are related to various CoMP-JT clustering choices. Note that we denote the set of $l_{\max}$ LiFi APs with the strongest RSS for UE u as L_u , which has $2^{l_{\max}} - 1$ nonempty subsets, and each of them corresponds to one CoMP-JT clustering strategy, i.e., $S_1, S_2, \dots$ or $S_{2^{l_{\max}}-1}$. For example, if we have $l_{\max} = 2$ and $L_u = \{1, 2\}$ for UE n , meaning that UE n receives the strongest signals from LiFi APs 1 and 2, then we have $\mathcal{S}_n = \{S_0, S_1, S_2, S_3\}$, where S_0, S_1, S_2 , and S_3 respectively denote the strategies that serve this UE by the WiFi AP, by LiFi AP 1, by LiFi AP 2, and by LiFi APs 1 and 2 in CoMP-JT mode. We denote the strategy chosen for UE u as B_u , while the values of α_u and β_u corresponding to B_u can be found via the lookup table.

- 3) Payoff Function $\{p_u(\mathbb{B})\}_{u \in U}$: With the objective of maximizing the system throughput under proportional fairness constraints, the payoff function for player u is defined as

$$p_u(\mathbb{B}) = \log(R_u) = \log[\eta_u^{\text{LiFi}} C_u^{\text{LiFi}} + \eta_u^{\text{WiFi}} C_u^{\text{WiFi}}], \quad (17)$$

where $\mathbb{B} = \{B_1, B_2, \dots, B_{|U|}\}$ represents the strategy profile of all players. Notably, the value of the payoff function is dependent on α and β as indicated in Eq. (15). ■

For the formulated game $\mathbb{G} = \{U, \{\mathcal{S}_u\}_{u \in U}, \{p_u(B_u)\}_{u \in U}\}$, a local decision function (LDF) is further defined as

$$\begin{aligned} v_u(B_{-u}, B_u) &= p_u(B_{-u}, B_u) \\ &+ \sum_{j \in T_u} [p_j(B_{-u}, B_u) - p_j(B_{-u}, \emptyset)], \end{aligned} \quad (18)$$

where B_{-u} denotes the strategy profile of all the UEs except u ; T_u denotes the set of neighbors of UE u , where we define that UE j is the neighbor of UE u if they have at least one common serving AP, i.e., $B_j \cap B_u \neq \emptyset$; and $p_j(B_{-u}, \emptyset)$ refers to the payoff of neighbor j as if player u gives up competing for the network resources, i.e., UE u would not connect with any AP.

Based on the constraint C2 in Eq. (16), we can know that (B_{-u}, B_u) would not be the solution to the formulated MINLP problem if $B_u = S_0$ and the required WiFi resources ratio $\eta_{\text{req}}^{\text{WiFi}} = \sum_{u \in U} \beta_u \eta_u^{\text{WiFi}}$ is larger than $\eta_{\max}^{\text{WiFi}}$. In this case, we normalize the allocated WiFi resources fraction as given by

$$\hat{\eta}_u^{\text{WiFi}} = \eta_u^{\text{WiFi}} \eta_{\max}^{\text{WiFi}} / \eta_{\text{req}}^{\text{WiFi}}, \quad \forall u \in U. \quad (19)$$

However, after the normalization, the QoE demands for the data rate of the UEs connecting to the WiFi AP could not be

satisfied. Hence, we redefine the LDF as

$$v_u(B_{-u}, S_0) = p_u(B_{-u}, S_0) + |T_u|\epsilon_u, \quad (20)$$

where $|\cdot|$ denotes the cardinality of a set and $\epsilon_u, \forall u \in \mathcal{U}$, represents to what extent the strategy choice of UE u influences the performances of the other users connected to the WiFi AP, whose value is randomly selected to enable Eq. (23) to hold. More explicitly, for larger ϵ_u , the value of $v_u(B_{-u}, S_0)$ becomes larger, and then the strategy S_0 is more likely to be selected, bringing larger performance influences to other UEs.

C. Nash Equilibrium and Best Response Dynamics

Naturally, for the formulated game model $\mathbb{G}$, it is expected to find the NE to obtain a stable solution. Next, we first prove the existence of the NE and then derive the NE as a suboptimal solution to the MINLP problem.

The NE, denoted as $\mathbb{B}^*$, is defined as the point at which no player can improve its utility by unilaterally changing its strategy [34], i.e.,

$$p_u(B_{-u}^*, B_u^*) \geq p_u(B_{-u}^*, B_u) \quad \forall B_u \in \mathcal{S}_u, \forall u \in \mathcal{U}. \quad (21)$$

Furthermore, the definitions of GOPG [34] is given as follows:

Definition 1 (GOPG): The game $\mathbb{G}$ is a GOPG if and only if a potential function $F(\mathbb{B})$ exists such that, $\forall u \in \mathcal{U}, \forall B_u', B_u \in \mathcal{S}_u$:

$$P(B_{-u}, B_u') - P(B_{-u}, B_u) > 0 \Rightarrow F_u(B_{-u}, B_u') - F_u(B_{-u}, B_u) > 0, \quad (22)$$

where $P(B_{-u}, B_u)$ is the sum of all players' utilities.

Based on the above discussions, we can further have the following Theorem 1:

Theorem 1: Game $\mathbb{G} = \{\mathcal{U}, \{\mathcal{S}_u\}_{u \in \mathcal{U}}, \{p_u(B_u)\}_{u \in \mathcal{U}}\}$ is a GOPG if it satisfies Eq. (23) and therefore admits at least one pure NE:

$$\epsilon_u \leq \log \frac{\eta_{\text{req}}^{\text{WiFi}} - \eta_u^{\text{WiFi}}}{\eta_{\text{req}}^{\text{WiFi}}}, \quad \forall u \in \mathcal{U}. \quad (23)$$

Proof 1: In the case $\eta_{\text{req}}^{\text{WiFi}} \leq \eta_{\text{max}}^{\text{WiFi}}$, there is no need to perform the normalization. According to the appendix in [17], we can derive $\forall B_u', B_u \in \mathcal{S}_u, \forall u \in \mathcal{U}$, and

$$v_u(B_{-u}, B_u') - v_u(B_{-u}, B_u) = P(B_{-u}, B_u') - P(B_{-u}, B_u), \quad (24)$$

which may enable Eq. (22) to hold.

Otherwise, when the strategy S_0 is selected and $\eta_{\text{req}}^{\text{WiFi}} > \eta_{\text{max}}^{\text{WiFi}}$ (without loss of generality, assume B_u is S_0), we denote the required WiFi resource ratios for strategy profiles $(B_{-u}, \emptyset)$ and (B_{-u}, S_0) as $\eta_{\text{req}}^{\text{WiFi}} - \eta_u^{\text{WiFi}}$ and $\eta_{\text{req}}^{\text{WiFi}}$, respectively.

Based on the appendix in [17], we can derive

$$\begin{aligned} P(B_{-u}, B_u') - P(B_{-u}, S_0) &= v_u(B_{-u}, B_u') \\ &- \left\{ p_u(B_{-u}, S_0) + \sum_{j \in T_u} [p_j(B_{-u}, S_0) - p_j(B_{-u}, \emptyset)] \right\} \\ &= v_u(B_{-u}, B_u') - \left\{ p_u(B_{-u}, S_0) + \sum_{j \in T_u} \left[\log \frac{\eta_{\text{max}}^{\text{WiFi}} \eta_u^{\text{WiFi}} C_j^{\text{WiFi}}}{\eta_{\text{req}}^{\text{WiFi}}} \right. \right. \\ &\quad \left. \left. - \log \frac{\eta_{\text{max}}^{\text{WiFi}} \eta_u^{\text{WiFi}} C_j^{\text{WiFi}}}{\eta_{\text{req}}^{\text{WiFi}} - \eta_u^{\text{WiFi}}} \right] \right\} \\ &= v_u(B_{-u}, B_u') - \left[p_u(B_{-u}, S_0) + \sum_{j \in T_u} \log \frac{\eta_{\text{req}}^{\text{WiFi}} - \eta_u^{\text{WiFi}}}{\eta_{\text{req}}^{\text{WiFi}}} \right] \\ &= v_u(B_{-u}, B_u') - \left[p_u(B_{-u}, S_0) + |T_u| \log \frac{\eta_{\text{req}}^{\text{WiFi}} - \eta_u^{\text{WiFi}}}{\eta_{\text{req}}^{\text{WiFi}}} \right]. \end{aligned} \quad (25)$$

If Eq. (23) holds, naturally we have

$$v_u(B_{-u}, B_u') - v_u(B_{-u}, B_u) \geq P(B_{-u}, B_u') - P(B_{-u}, B_u), \quad (26)$$

which means that Eq. (22) also holds.

Therefore, we always have Eq. (24) for all possible $\eta_{\text{req}}^{\text{WiFi}}$ if Eq. (23) holds, meaning that the formulated game model is a GOPG and thereby admits at least one NE due to the finite improvement property [34]. Notably, there exist multiple NEs for the GOPG model [17]. The proof is completed. ■

Subsequently, we define the best response of player u to the strategy profile B_{-u} as strategy ζ_u such that

$$\zeta_u = \arg \max_{B_u \in \mathcal{S}_u} v_u(B_{-u}, B_u). \quad (27)$$

By sequentially choosing the best response for each player $u \in \mathcal{U}$, the BR dynamics [17] will converge to a NE [34].

Concerning the resource allocations based on Eq. (14), the resources of some LiFi APs may not be fully occupied at the NE found based on the BR dynamic, i.e., the resource occupying ratio $\varepsilon_l = \sum_{u \in \mathcal{U}} \alpha_{u,l} \eta_u^{\text{LiFi}} < 1, l \in \mathcal{L}$. For each of these LiFi APs, we will further allocate the remaining resources uniformly to the UEs that only connect to the LiFi AP.

Thus, we can obtain the joint user-centric LiFi CoMP-JT clustering and LB algorithm as Algorithm 1, whose computation complexity in each iteration is $O(2^{l_{\text{max}}} |\mathcal{U}|)$. Inspired by the research works presented by [35], we propose to apply game theory to provide insight on how uncoordinated UE and APs should associate. Furthermore, since the decision variables are binary, the rate maximization problem formulated as Eq. (16) is a nonconvex problem, and thus the computational complexity is high to find out the optimal UE-AP association. Besides, even if the optimal UE-AP association could be found, since the cluster is dynamically allocated, due to high complexity of solving Eq. (16), the UE-AP association could not be updated in real time, thereby leading to impractical solutions. Accordingly, based on game theory, we analyze the strategic situations of UEs and the AP, and then we derive the optimal

Algorithm 1. Joint LiFi CoMP-JT Clustering and LB Algorithm

Require: UEs' CSI and QoE requirements;
Ensure: Association indicators and time resource allocation;

- 1: Initialize α , β , and $\{S_u\}_{u \in U}$;
- 2: **while** it has not converged **do**
- 3: **for** each UE $u \in U$ **do**
- 4: Set the value of ϵ_u to enable Eq. (23);
- 5: **for** each strategy $S_i \in S_u$ **do**
- 6: Calculate $v_u(B_{-u}, S_i)$ via Eqs. (17)–(20);
- 7: Choose ζ_u for UE u ;
- 8: Calculate $\varepsilon_l \forall l \in L$;
- 9: **for** the LiFi AP with $\varepsilon_l < 1$ **do**
- 10: Allocate the left resources uniformly to UEs without JT operation.

strategy they should select to maximize their own utility. Thus, their strategy can adapt to the dynamically changing network topology in real time, which greatly enhances the practicality of the proposed design.

4. SIMULATION RESULTS AND ANALYSES

In the simulations, as shown in Fig. 1, a $10 \text{ m} \times 10 \text{ m} \times 3 \text{ m}$ room with a 4×4 LiFi AP matrix in UFR deployment and a WiFi AP on the ceiling is employed. Tables 2 and 3 summarize the simulation parameters for the LiFi and WiFi channels, respectively. The UEs are distributed randomly at the height of 1 m, and different types of services occur with the same probability. The LiFi detectors are all facing upwards. In order to achieve a better trade-off between time resource requirements and the user satisfaction degree, the minimum data rate requirements are set as 0.9 for all UEs, i.e., $Q_{u,i} = 0.9, \forall u \in U$. Furthermore, the symbol error probability P_e is set as 10^{-2} [22], while the modulation order is set as $M = 4$ for both LiFi and WiFi subsystems.

Notably, in the special case that some UEs are occasionally located in the corner or obstructed by obstacles, due to the LoS propagation characteristics of VL bands, the RSS is too weak to perform CoMP. Hence, these UEs would be allocated to the WiFi AP or one LiFi AP and served via SPT. In addition, as mentioned above, at the beginning of each time interval, the UEs estimate the CSI for both LiFi and WiFi links and report the CSI via uplinks to the CCU [6].

Table 2. Simulation Parameters for LiFi Channels

Parameters	Value
Gain of optical filter	1
Physical area of detector	1 cm^2
Field of view of detector	120°
Half-intensity radiation angle	60°
Optical-to-electric power conversion coefficient	3
Detector responsivity	0.53 A/W
PSD of LiFi noise	$10^{-21} \text{ A}^2/\text{W}$
LiFi transmit power	6 W
Bandwidth per LiFi AP	40 MHz

Table 3. Simulation Parameters for WiFi Channels

Parameter	Value
Breakpoint distance	5 m
Shadow fading standard deviation before breakpoint	3 dB
Shadow fading standard deviation after breakpoint	5 dB
Central frequency	2.4 GHz
Arrival angle of LOS path	45°
Downlink resources quota	0.8
WiFi transmit power	20 dBm
PSD of RF noise	-135 dBm/Hz
Bandwidth per WiFi AP	20 MHz

For performance comparisons, four benchmarks are considered: the minimum distance (Dmin) strategy that always connects the UE to the nearest AP; the maximum received signal strength (Smax) strategy that always connects the UE to the AP with the highest signal-to-noise ratio (SNR); and two schemes in [5] termed as LB: UFR and LB: CT-2, where the LBs are respectively presented in the UFR and static two-cell CT deployment.

It is worth pointing out that no precoding modules are employed in any of these four benchmark systems and in our proposed system. Besides, only the Dmin aided system does not need to know the CSI, but it still requires additional signaling to learn the position information. The other three benchmark schemes as well as our proposed system need to evaluate the SNR or the signal strength for resource allocations [17]. Since both the signaling overhead and hardware cost of benchmark schemes and our design are similar, the fairness of performance comparisons among benchmark schemes and our design provided as follows can be guaranteed.

A. Convergence Performance

Figure 3 provides the numerical analysis of the convergence in the iterations. It can be seen that when there are 80 UEs and the number of iterations increases, the value of the objective function approaches a specific threshold, which demonstrates the convergence of iterations. In addition, according to Theorem 1, since the formulated game is a GOPG, the best response dynamic has been proved to be convergent to the desired NE [17,34]. Based on the numerical analysis and the theoretical analysis in [17,34], we can come to the conclusion that the proposed design has the property of convergence.

B. System Throughput Performances

Figure 4 compares the throughput of hybrid networks using different schemes. It can be seen that the systems using the presented design exhibit the highest throughput. Thanks to the efficient load management, our proposed scheme outperforms the Smax and Dmin schemes, especially when there is a great number of UEs. Besides, the proposed scheme outperforms the LB: UFR [5] scheme since the LiFi ICI is suppressed by the CoMP-JT schemes, enabling the LiFi APs to better serve the cell-edge UEs. Moreover, higher throughput can be achieved by our design than the LB: CT-2 [5] scheme, wherein the

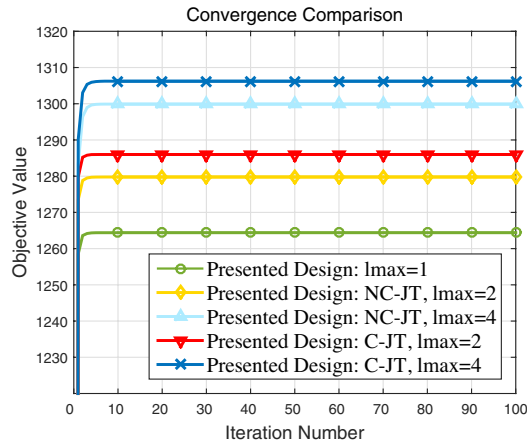


Fig. 3. Numerical analysis of the iteration convergence when there are 80 UEs.

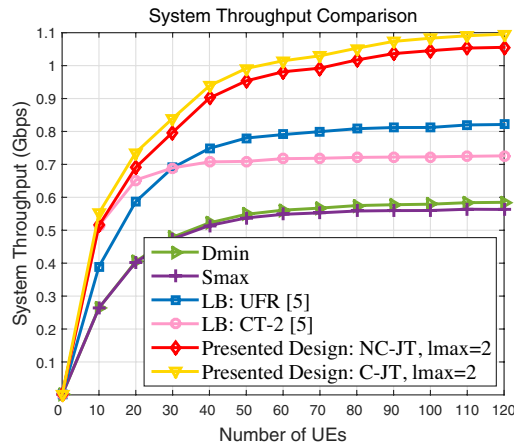


Fig. 4. System throughput versus the number of UEs.

AP clustering is static and network-centric, and thus the system suffers from inter-cluster interference. In addition, we can observe that the C-JT provides better performance than NC-JT, which is consistent with the conclusion given in [16].

C. Evaluation of User Satisfaction

To evaluate the user satisfaction level in hybrid networks, Fig. 5 compares the user satisfaction degree, which is defined as the ratio of UEs whose QoE demands on data rate are satisfied. We can see that with the aid of our design, many more UEs are satisfied in the hybrid system. The reason is that our proposed design is QoE-aware, namely, the WiFi resource is allocated to meet the minimum data rate demand of the UEs. By contrast, the benchmark schemes do not take the UEs' QoE demands into account in the design.

Figure 6 compares the cumulative distribution function (CDF) of the UE QoE index in the hybrid system applying different schemes when there are 80 UEs. We can observe that the UE QoE indexes achieved by our design attain more satisfactory distribution than those achieved by the counterpart schemes. More explicitly, in our proposed system, there are many more UEs whose QoE index is higher than 0.9, while

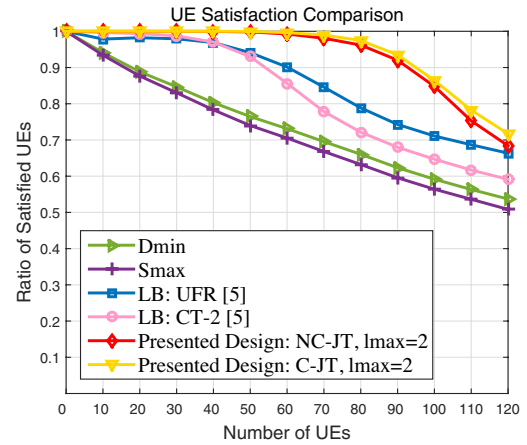


Fig. 5. User satisfied with the data rate versus the number of UEs.

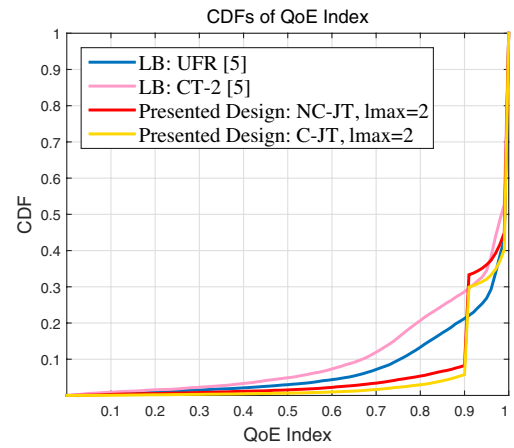


Fig. 6. CDFs of the QoE index when there are 80 UEs.

there are many fewer UEs whose QoE index is lower than 0.8. From the different utility functions adopted in WiFi or LiFi networks for time resource allocation, i.e., Eq. (13) and Eq. (14), we can know that the WiFi resources are allocated to meet the UEs' QoE demands based on the minimum data rate, while LiFi resources are allocated to make the best efforts to provide high-speed links. As a result, the UEs connected to the WiFi network will achieve the required minimum data rate corresponding to the predefined satisfaction level of $Q_{u,i}$, i.e., 0.9 as mentioned above. Thus, in Fig. 6, the CDFs of the QoE index of hybrid systems applying our proposed design exhibit a step-like improvement at 0.9.

D. SINR and Outage Performances

In addition to the user demands of the data rate, we also take the reliability performances into account. The reliability performances are mainly determined by the bit error rate (BER), while the BER is dependent on the SINR and the modulation order. Since at the stage of establishing the UE-AP association, no adaptive coding and modulation schemes can be applied, the SINR will be determinant for the reliability performances. In Fig. 7, we provide the simulation results to demonstrate that the average SINR could be effectively improved by our

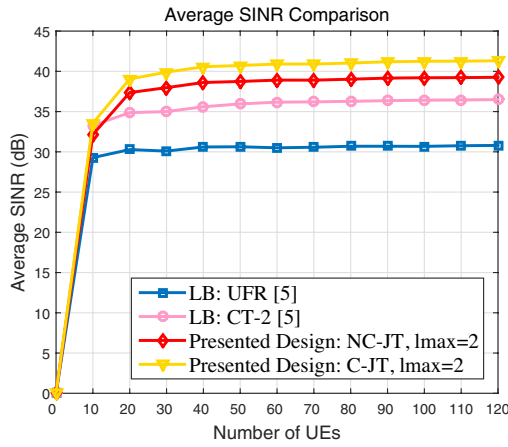


Fig. 7. Average SINR versus the number of UEs.

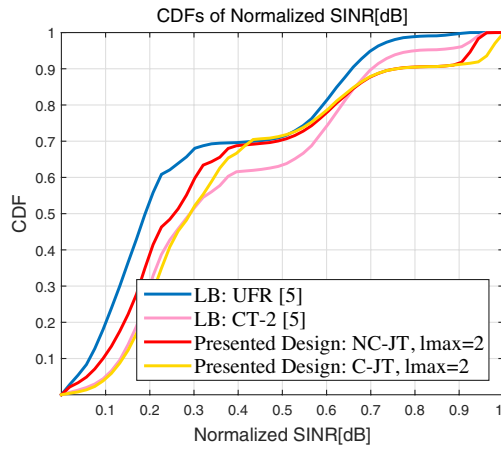


Fig. 8. CDFs of the normalized SINR [dB] when there are 80 UEs.

design. In addition, Fig. 8 evaluates the CDFs of normalized SINR [decibels (dB)] in hybrid systems applying different schemes when there are 80 UEs, from which we can observe that the SINR follows a more satisfactory distribution in our design aided systems. Thus, with the enhanced SINR, the reliability would accordingly be improved when the physical layer configurations remain the same.

Moreover, Fig. 9 compares the outage rate, which is defined as the ratio of UEs whose SINR value is below a predefined threshold $SINR_{th}$. As shown in the figure, no matter how many UEs are in the indoor scenario, the CoMP-JT schemes, including our presented schemes and the LB: CT-2 [5] scheme, can achieve a lower outage rate than a non-CoMP-JT scheme, i.e., the LB: UFR [5] scheme. The reason is that the management of the LiFi ICI would improve the SINR of the cell-edge UEs. Besides, three CoMP-JT schemes achieve similar outage rate performances when there is a large number of UEs in the hybrid system. The LB: CT-2 [5] scheme slightly outperforms our design using the NC-JT, because the LB: CT-2 [5] scheme performs the CoMP-JT for every UE connected to the LiFi network. Additionally, our proposed design using the C-JT obviously outperforms the other two schemes using the

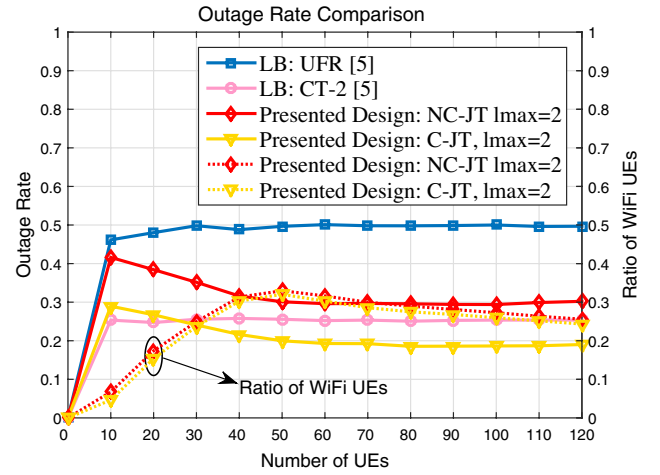


Fig. 9. Outage rate versus the number of UEs where $SINR_{th} = 10$ dB.

NC-JT, since the C-JT exhibits advantages in SINR improvement over the NC-JT scheme.

It is also worth mentioning that the outage rate is not monotonically increasing as the number of UEs increases. The main reason is that, due to the constrained time resource, the WiFi AP can serve a limited number of users, and thus when UEs consume the available time resources, no more UEs are allowed to access the WiFi AP. Accordingly, as shown in Fig. 9, as the number of UEs increases from 20 to 80, the ratio of WiFi UEs increases, which would contribute to a higher overall SINR improvement in the LiFi subsystem (and thus a lower outage rate), since the UEs allocated to the WiFi AP usually have worse LiFi channel conditions. However, as the occupied resources reach the fraction of η_{max}^{WiFi} , the WiFi AP cannot serve more UEs because it needs to guarantee the minimum data rate to satisfy the associated UEs' QoE requirements. Therefore, the ratio of WiFi UEs slightly decreases after the number of UEs reaches 80, leading to the slight decrease of overall SINR improvement in the LiFi subsystem (and thus the increase of outage rate).

E. Performances with Different l_{max}

Finally, Figs. 10 and 11 evaluate the performances of our proposed system with different l_{max} in the case of 80 UEs. In Fig. 10, we can see that the system throughput increases with larger l_{max} and becomes saturated when l_{max} is larger than 4. The reason is that when the allowed cluster size increases, the UE's SINR will be improved. However, as indicated in Eq. (16), any cluster decision with larger cluster size is only allowed if it contributes to the maximization of the system throughput under proportional fairness constraints; therefore, the throughput will become saturated when l_{max} increases to some threshold value. Moreover, it is worth mentioning that the complexity of our proposed algorithm and that of UE scheduling also increases with l_{max} , and hence we need to select l_{max} carefully to achieve a better trade-off between the throughput and the complexity. In the special case that $l_{max} = 1$, all UEs are served via the SPT scheme, and hence the system throughput is lower than that using LB: UFR [5] due

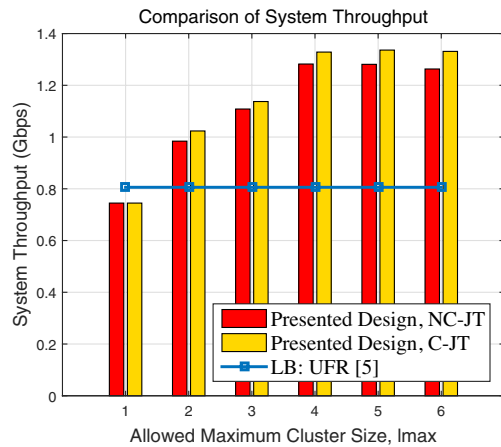


Fig. 10. System throughput versus $l_{\max}$ when there are 80 UEs.

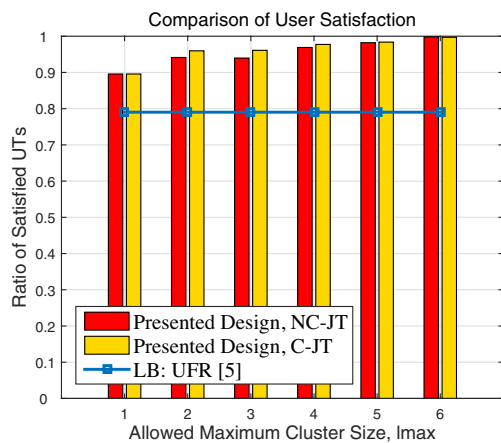


Fig. 11. Ratio of users satisfied with the data rate versus $l_{\max}$ when there are 80 UEs.

to the suboptimality. For all other $l_{\max} > 1$, better throughput performances can be attained thanks to the proposed joint clustering and LB design.

In Fig. 11, we can notice that our design always outperforms the LB: UFR [5] scheme in terms of the ratio of users satisfied with the data rate, thanks to our design that the WiFi AP is also employed to meet the minimum data rate requirements of UEs across the whole indoor environment. Besides, we could observe from the figure that the ratio of users satisfied with the data rate does not increase with larger $l_{\max}$, since the clustering strategy is mainly employed to improve the throughput performances and thus has limited contributions to meet the minimum data rate demands for users.

5. CONCLUSION

In this paper, we propose a joint user-centric LiFi CoMP-JT clustering and LB design to simultaneously address both the imbalanced cell load and the LiFi ICI issues for hybrid LiFi and WiFi networks. We take the mutual influence of the LiFi CoMP-JT clustering and LB on performances into account in our design, and we formulate the joint design as a MINLP problem. With the aim of maximizing the system throughput

under proportional fairness constraints, we build a GOPG model to obtain the suboptimal solution for efficient UE-AP association and time slot resource allocations. Simulation results verify the effectiveness of our proposed design in terms of the system throughput, the UE satisfaction degree, and the outage rate. Thus, thanks to the proposed joint user-centric LiFi CoMP-JT and LB design, the data rate can be improved for systems with the same physical layer configurations in indoor communication networks. In future research work, we would further investigate the cross-layer optimization of the clustering and the LB strategy as well as the adaptive coding and modulation scheme to achieve better transmission quality. Besides, the research works could be further extended to outdoor scenarios, wherein the mobility of UEs would be taken into account in the design.

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